Electrical AC BOM (Phase 1)

As-of date: `2026-07-26`

Purpose: maintain the purchased Phase 1 AC architecture baseline: portable `30A` EMS, `30A` shore inlet/cord, one `6-way` DIN enclosure, `30A` AC-in breaker, `30A` AC-out main breaker, and two active `20A` GFCI-protected AC-out branches. Final physical outlet locations and enclosure access remain measurement-gated in the installed camper.

Related docs:

- `docs/implementation/ELECTRICAL_overview_diagram.md`
- `docs/core/SYSTEMS.md`
- `docs/core/TRACKING.md`
- `bom/bom_estimated_items.csv`

Historical provenance

- INSTALL_MINUS_12_READINESS_PLAN preserves the May 7 install-window AC planning context; this file owns current Phase 1 AC procurement/implementation.

Physical shore-inlet install gate (`2026-07-19`)

- The L5-30 inlet remains purchased but not yet cut into the camper/bed-side interface.
- The electrical module is now hard-mounted. The MultiPlus and combined AC breaker enclosure were removed for the lift and must be remounted into the backer's embedded pronged/spiked T-nuts before the final shore path is terminated.
- From the remounted AC-in breaker endpoint, mock the full `10/3` path to the candidate exterior inlet with required bend radius, cable support, drip/water management, service loop, strain relief, and access to both sides of the cut.
- Any small cable opening through the plywood electrical backer requires a correctly sized grommet/bushing or gland plus independent cable support/strain relief; bare `10/3` must not bear on a raw plywood edge.
- Confirm the exterior cover swing, connector clearance, wall/backing thickness, hidden structure/no-drill zone, butyl bedding land, finish-seal geometry, and a path that does not share the wet-side chase.
- Cut only after the complete inside and outside route is proven. Do not let the easiest exterior location create an inaccessible cable entry or service-obscuring bend behind the electrical module.

Locked AC Architecture

AC-in chain (shore to inverter)

- `shore source/adapters -> portable 30A EMS -> 30A shore cord -> L5-30 shore inlet -> combined 6-way AC DIN enclosure -> 30A UL489 AC-in breaker/disconnect -> MultiPlus AC-in (L/N/PE)`
- AC-in conductors are `10 AWG` / `10/3` on the protected AC-in path (`30A` hardware basis).
- MultiPlus input current limit is set to actual source when adapters are used. Use `10A` for first household tests and `12A` maximum policy on a normal `15A` outlet; do not leave the limit at `50A` on adapter/household shore.

AC-out chain (inverter-backed branch distribution)

- `MultiPlus AC-out-1 -> 10/3 feeder -> combined 6-way AC DIN enclosure -> 30A UL489 AC-out main breaker -> 20A Branch A + 20A Branch B -> GFCI receptacle per branch`
- Current purchased receptacle plan is `2` active GFCI receptacles total: Branch A = office/driver side, Branch B = galley/passenger side. Prior downstream non-GFCI receptacle locations are obsolete unless a later layout revision reopens them.
- AC-out branch hardware is not required to perform the initial AC-in-only battery charging test, but it is now included in the purchased Phase 1 cart.

Neutral and ground handling

- AC-in and AC-out neutral termination paths remain isolated.
- Separate AC-in neutral pass-through/termination and AC-out neutral bar are required inside the combined enclosure.
- Common equipment grounding/PE bus is acceptable; do not use it as a neutral.
- Continuous equipment ground path and chassis bond are required end-to-end.
- Do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring.

AC-out-2 policy

- `AC-out-2` is **reserve-only** in Phase 1.
- Keep labeled panel space and capped route only; no energized branch hardware is procured for this path in Phase 1.

Required Purchasable Components (Phase 1)

Immediate AC-in-only initial charge path

These rows unblock safe MultiPlus shore charging and should not wait on final receptacle count:

Component class	Qty	Rating/listing requirement	BOM row(s)	Phase 1 status
Shore inlet	1	30A 125V L5-30 shore inlet with exterior cover	107	Purchased
Shore cord + household adapter	1 cord + 1 dogbone	30A TT-30P-to-L5-30R cord plus 15A household dogbone adapter	108	Purchased
Portable EMS/surge protector	1	Portable 30A EMS with open-neutral/polarity/voltage protection	123	Purchased
AC-in breaker/disconnect	1 of 2	DIN-mount UL 489 1-pole 30A 120VAC	13	Purchased
Combined AC DIN enclosure and bus hardware	1 enclosure + accessories	6-way DIN enclosure; physically isolate AC-in and AC-out neutrals	109, 14	Purchased
Shore + AC-in feed cable	as routed	10/3 stranded tinned-copper cable for C-28/C-29 (30A path)	114	Purchased
Shore-inlet seal/bedding consumables	per install	Butyl bedding + compatible exterior polyurethane perimeter/finish seal	179, 180	Purchased
Strain relief/grommets/clamps/labels/ferrules	per entries	Sized for selected cable/enclosure terminals	38, 41, 43, 44, 45	Cable glands on hand; ferrule kit purchased

Full AC-out distribution / final Phase 1 branch hardware

Component class	Qty	Rating/listing requirement	BOM row(s)	Phase 1 status
Shore inlet	1	30A 125V L5-30 shore inlet with exterior cover	107	Purchased
Shore cord + household adapter	1 cord + 1 dogbone	30A TT-30P-to-L5-30R cord plus 15A household dogbone adapter	108	Purchased
Portable EMS/surge protector	1	Portable 30A EMS used source-side before the shore cord	123	Purchased
AC-in breaker/disconnect	1	DIN-mount UL 489 1-pole 30A 120VAC	13	Purchased
AC-out main breaker	1	DIN-mount UL 489 1-pole 30A 120VAC, ahead of branch breakers	13	Purchased
Combined DIN enclosure	1	6-way DIN enclosure with slot plan: 30A in, 30A out, 20A, 20A, 2x blank/spare	109	Purchased
AC-out branch breaker set	2 active	DIN-mount UL 489 branch breakers: 20A + 20A	110	Purchased
DIN accessory kit	1 kit	Neutral isolation hardware, PE/ground bus support, ferrules, labels/blanks as needed	14, 41	Purchased / labels-blanks confirm on hand
GFCI receptacles	2	20A self-test GFCI receptacles, one per active branch	15	Purchased
Standard downstream receptacles	0 in current scope	Prior downstream non-GFCI receptacle plan is obsolete	111	Obsolete

Outlet boxes + covers/faceplates + clamps	2 sets	Two active GFCI receptacles have covers/plates; final boxes/clamps are install-fit details using on-hand hardware as needed	112	Install-fit detail / not a first-charge blocker
AC branch cable	30 ft purchased	12/3 stranded triplex branch cable (C-31/C-32)	113	Purchased
Shore + AC-in/AC-out feeder cable	20 ft purchased	10/3 stranded triplex for C-28/C-29/C-30 (30A paths)	114	Purchased
Strain relief/cable glands	per enclosure entries	Use assorted on-hand entry hardware sized during physical layout	44	On-hand fitment stock
Grommets	per pass-through points	Abrasion protection at penetrations, selected hands-on during routing	43	On-hand / install fit
P-clamps and retention hardware	per route	Cable support and vibration control, selected hands-on during routing	45	On-hand / install fit
Loom/sleeving	per exposed runs	Harness abrasion protection	42	Required
Heat shrink (adhesive)	install consumable	Termination sealing and strain relief support	38	Required
Ferrules/terminals (AC-relevant)	install consumable	Sized to 10 AWG and 12 AWG terminations as required by device terminals	41, 116	Required
AC-out-2 branch breaker/protection hardware	0 in Phase 1	Reserve-only route, no energized branch hardware in this phase	N/A	Reserve-only (not procured)

First-Live AC-In Test Result (`2026-05-27`)

- AC Input 1 should be labeled `Shore power` for this mobile/source-current-limited system.
- MultiPlus switch `II` is charger-only; switch `I` is inverter/charger normal mode; `0` is off.
- Short shore test passed with household-source current limiting: about `1294W` shore input and about `54.3V x 21.6A` (`~1173W`) battery charge in bulk.
- Current disconnect practice: connect RV/EMS/load side first, then energize shore; disconnect in reverse by de-energizing/unplugging shore source before disconnecting the RV side.
- This result does not close battery-charge-profile commissioning. Program/verify the MultiPlus lithium settings before sustained charging.

Manual AC Validation Checklist

0) AC-in-only initial charger validation

- Confirm AC-in physical order: `shore source/adapters -> portable EMS -> shore cord -> inlet -> AC-in breaker/disconnect -> MultiPlus AC-in`.
- Confirm AC-out breakers/loads are disconnected or not yet installed for the first battery-charge test.
- Confirm MultiPlus input current limit is set to actual source (`10A` first household test, `12A` max on normal `15A`, actual rating for `20A` / `30A`).
- Confirm AC Input 1 is labeled/configured as `Shore power`.
- Confirm battery charge profile is intentionally set for the selected LiFePO4 voltage basis before sustained charging.
- Confirm AC-in and AC-out neutral paths are not mixed.

Use this checklist as the continuing acceptance gate before sustained shore charging, AC-out branch energization, or any AC enclosure closeout.

1) Topology integrity

- Confirm one unique AC-in chain exists: `shore source/adapters -> portable EMS -> cord -> inlet -> AC-in breaker -> MultiPlus AC-in`.
- Confirm one unique AC-out-1 chain exists: `MultiPlus AC-out-1 -> 30A AC-out main -> 20A branch breakers -> GFCI receptacles`.
- Confirm `AC-out-2` is documented as reserve-only and not active in Phase 1 procurement.
- Confirm current receptacle count is locked to `2` active GFCI receptacles unless a later layout revision reopens downstream receptacles.

2) Protection coordination

- Confirm AC-in breaker is `30A` and AC-in conductors are `10 AWG`.
- Confirm AC-out main breaker is `30A` and the MultiPlus-to-enclosure feeder is `10 AWG`.
- Confirm branch OCP values are `20A` and `20A` with `12 AWG` branch conductors.
- Confirm breaker listing basis is `UL 489` (or equivalent NRTL listing) for branch/feeder use.

3) Neutral/ground correctness

- Confirm AC-in and AC-out neutral paths are isolated.
- Confirm input and output neutral paths are separately landed/passed through inside the combined enclosure.
- Confirm the equipment grounding/PE path is common and continuous.
- Confirm continuous equipment grounding and chassis bond are documented.
- Confirm no fixed downstream neutral-ground bond is added in branch receptacle wiring.

4) Procurement completeness

- Confirm every required AC component class has a BOM row mapping.
- Confirm no AC-critical component exists only as implied text.
- Confirm AC-out-2 hardware remains excluded from Phase 1 carts.

5) Documentation parity

- Confirm AC assumptions match across:
  - `docs/implementation/ELECTRICAL_AC_BOM.md`
  - `docs/implementation/ELECTRICAL_overview_diagram.md`
  - `docs/core/SYSTEMS.md`
  - `docs/core/TRACKING.md`
  - `bom/bom_estimated_items.csv`

6) Operating scenarios

- AC-in-only initial charge: no AC-out loads connected, MultiPlus charger behavior documented.
- Shore present (`30A` source): pass-through + charging behavior documented.
- Shore present via `15A/20A` adapter: current-limit setting policy documented.
- Shore absent: inverter-backed `AC-out-1` behavior documented.
- GFCI trip/reset behavior per branch is called out for commissioning test.

Procurement Notes

- DIN rail is a mounting method; breaker listing and rating remain the controlling requirement.
- Lowest-cost listed policy is acceptable only if each selected device has verifiable NRTL listing (`UL` or `ETL`) for intended use.
- Purchased SKU lock is recorded in `bom/bom_estimated_items.csv` for rows `13`, `327`, `14`, `15`, `41`, `107`, `108`, `109`, `110`, `330`, `113`, `114`, `123`, `179`, and `180`. Rows `13` / `327` are the AC-in/AC-out `30A` pair; rows `110` / `330` are the two `20A` branch breakers. Row `181` was same-order tire-deflator/off-road support hardware and is intentionally outside AC scope.
- Inactive BOM row `111` preserves the obsolete downstream-receptacle concept. Active row `112` is only an install-fit box/cover/clamp detail around the two active GFCIs, not an AC design blocker.
- Current utilization note (`2026-05-16`): AC protection chain and purchase scope are locked for a `30A` system with two active `20A` branches. Do not add a third active AC branch without revisiting the AC-out main/enclosure/feed plan.